

Complete Dataset Evaluation

Sets A through F — All Results

EE 123 Project — Ved, Ching, Erick

Set A: source-vctk1k (1000 real, trained on)

Set B: target-vctk1k (1000 real, NOT trained on)

Set C: converted-vctk1k (1000 spoof, trained on)

Set D: source-vctk500 (500 real, NOT trained on)

Set E: target-vctk500 (500 real, NOT trained on)

Set F: converted-vctk500 (500 spoof, NOT trained on)

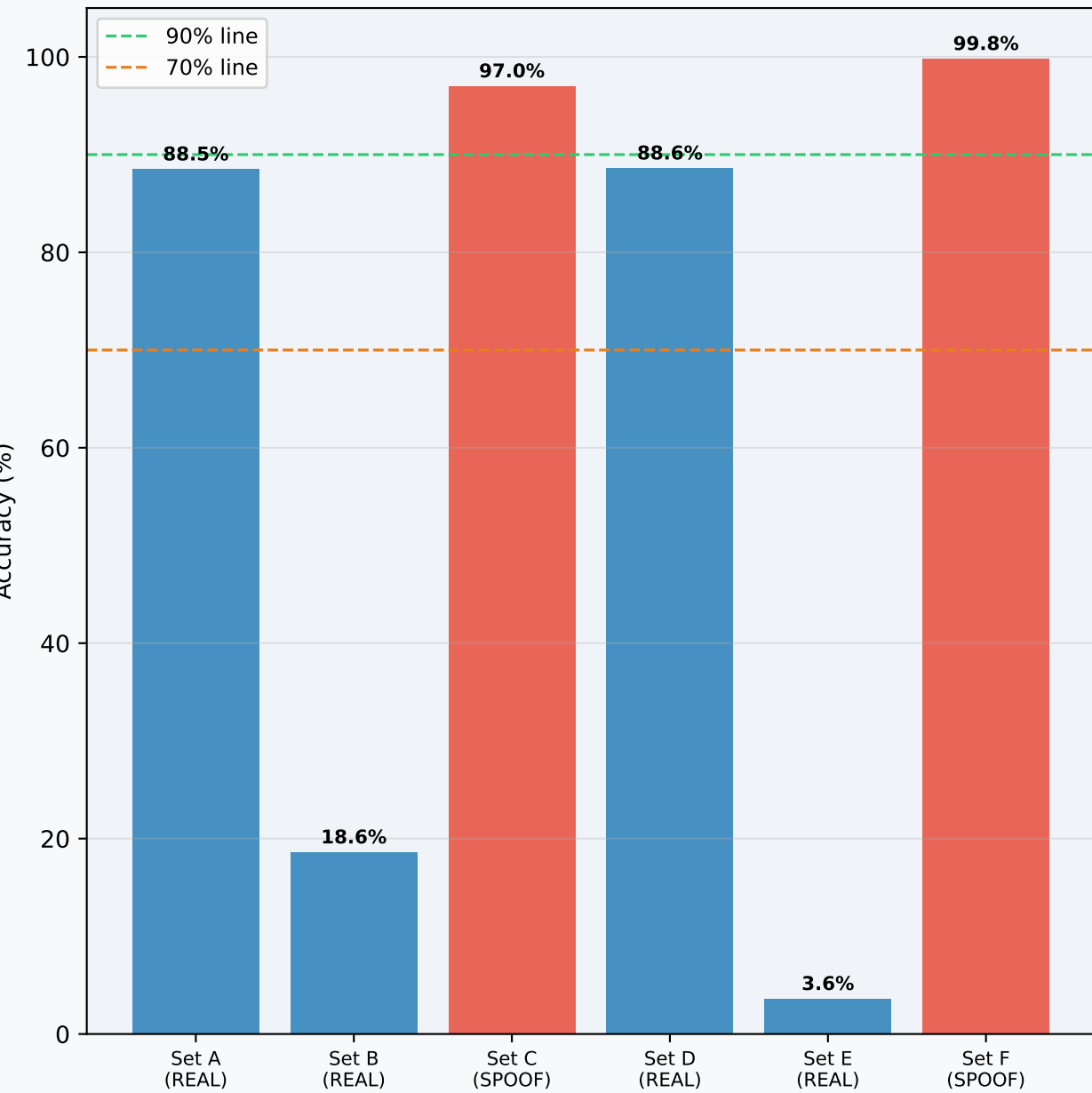
Full Dataset Evaluation — Sets A through F

Set	Folder	True Label	Training?	Total	LR Correct	LR%	XGB Correct	XGB%	AVG Correct	AVG%
Set A	source-vctk1k	REAL	trained on	1000	885/1000	88.5%	990/1000	99.0%	929/1000	92.9%
Set B	target-vctk1k	REAL	NOT trained on	1000	186/1000	18.6%	461/1000	46.1%	186/1000	18.6%
Set C	converted-vctk1k	SPOOF	trained on	1000	970/1000	97.0%	992/1000	99.2%	996/1000	99.6%
Set D	source-vctk500	REAL	NOT trained on	500	443/500	88.6%	452/500	90.4%	449/500	89.8%
Set E	target-vctk500	REAL	NOT trained on	500	18/500	3.6%	305/500	61.0%	19/500	3.8%
Set F	converted-vctk500	SPOOF	NOT trained on	500	499/500	99.8%	455/500	91.0%	498/500	99.6%

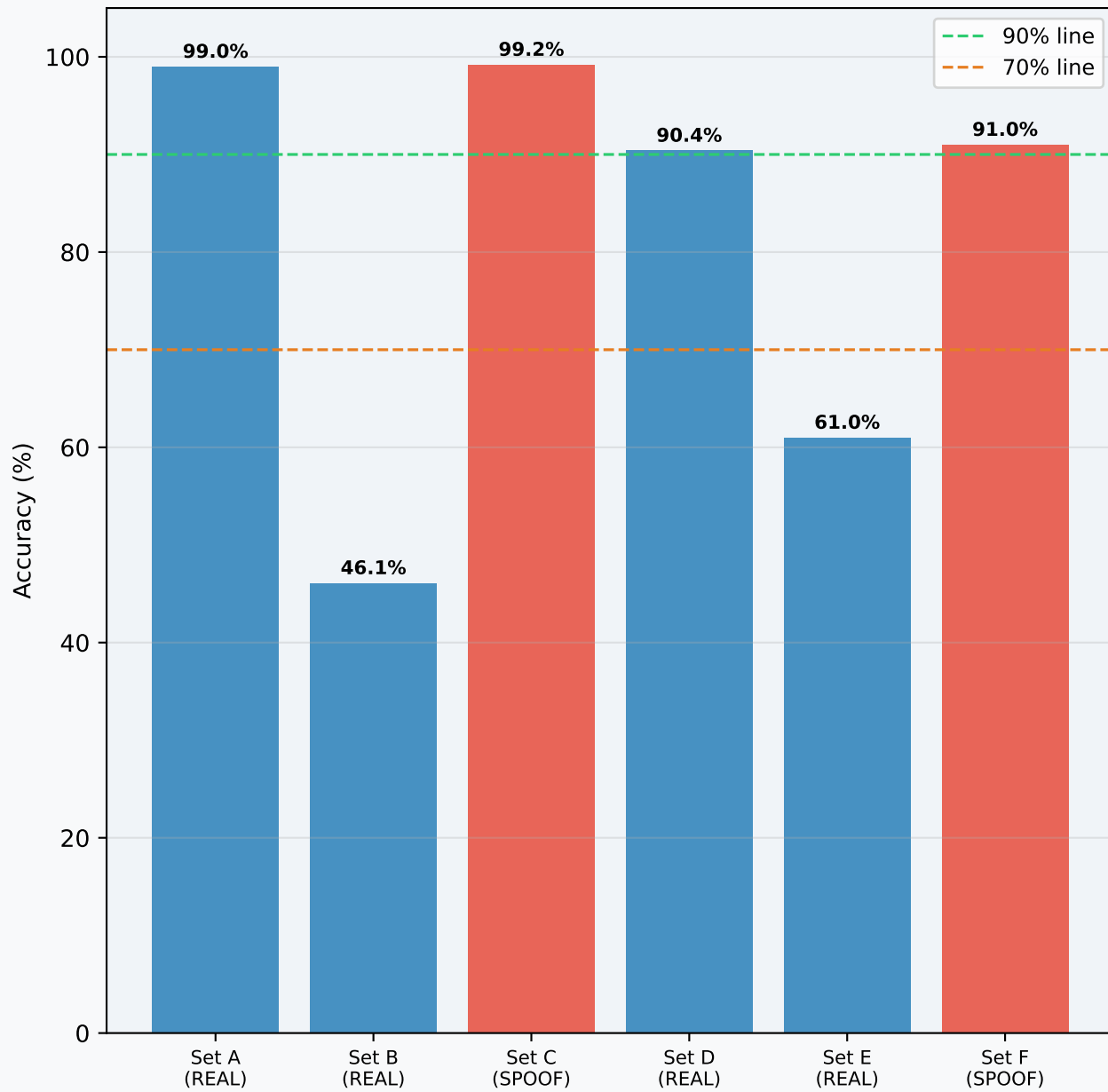
Green = ≥ 90% accuracy | Orange = 70-90% | Red = < 70%
Blue rows = real audio | Red rows = spoof audio

Accuracy Across All 6 Sets by Model

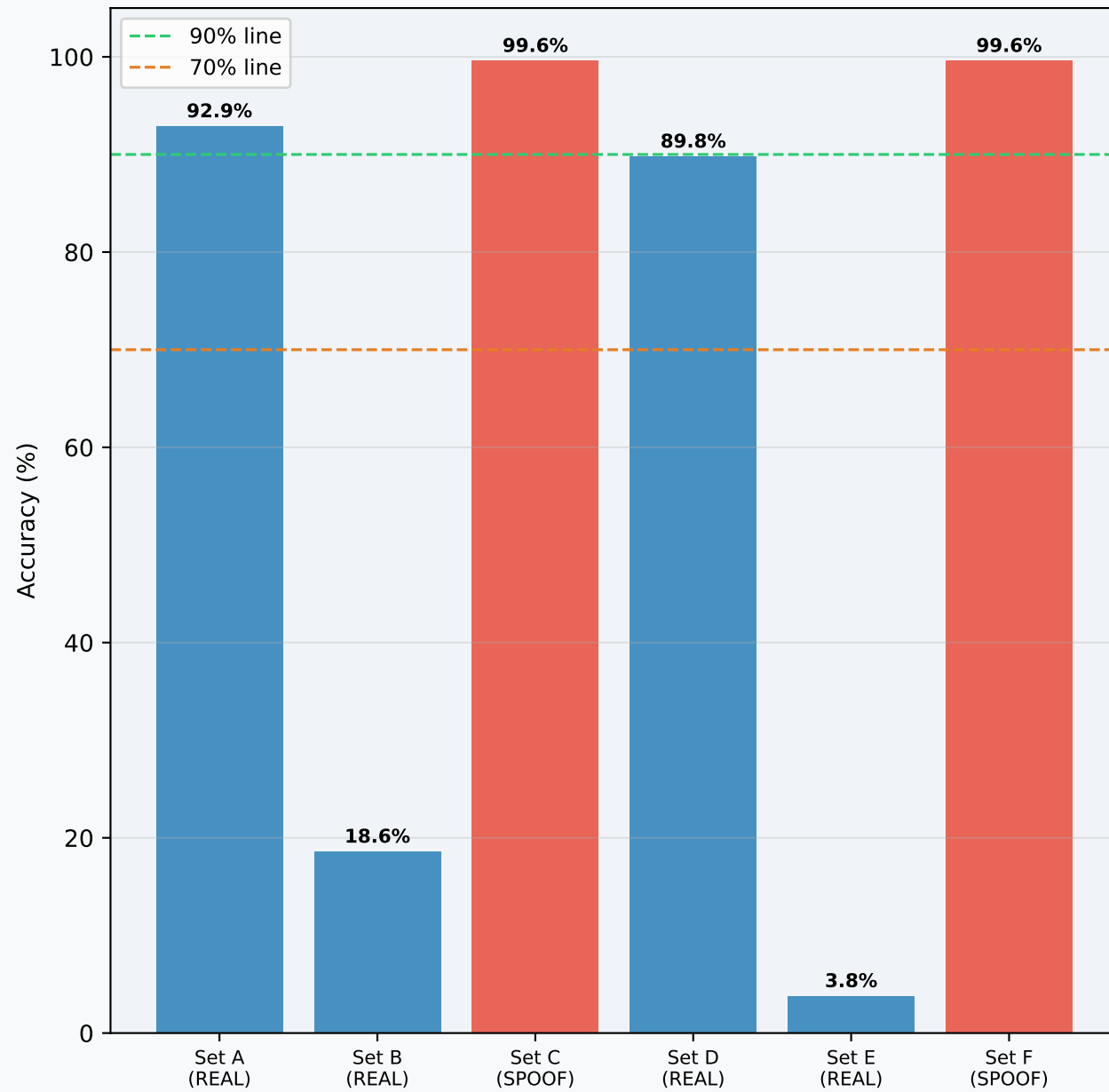
LR Accuracy per Set



XGBoost Accuracy per Set



AVG Ensemble Accuracy per Set

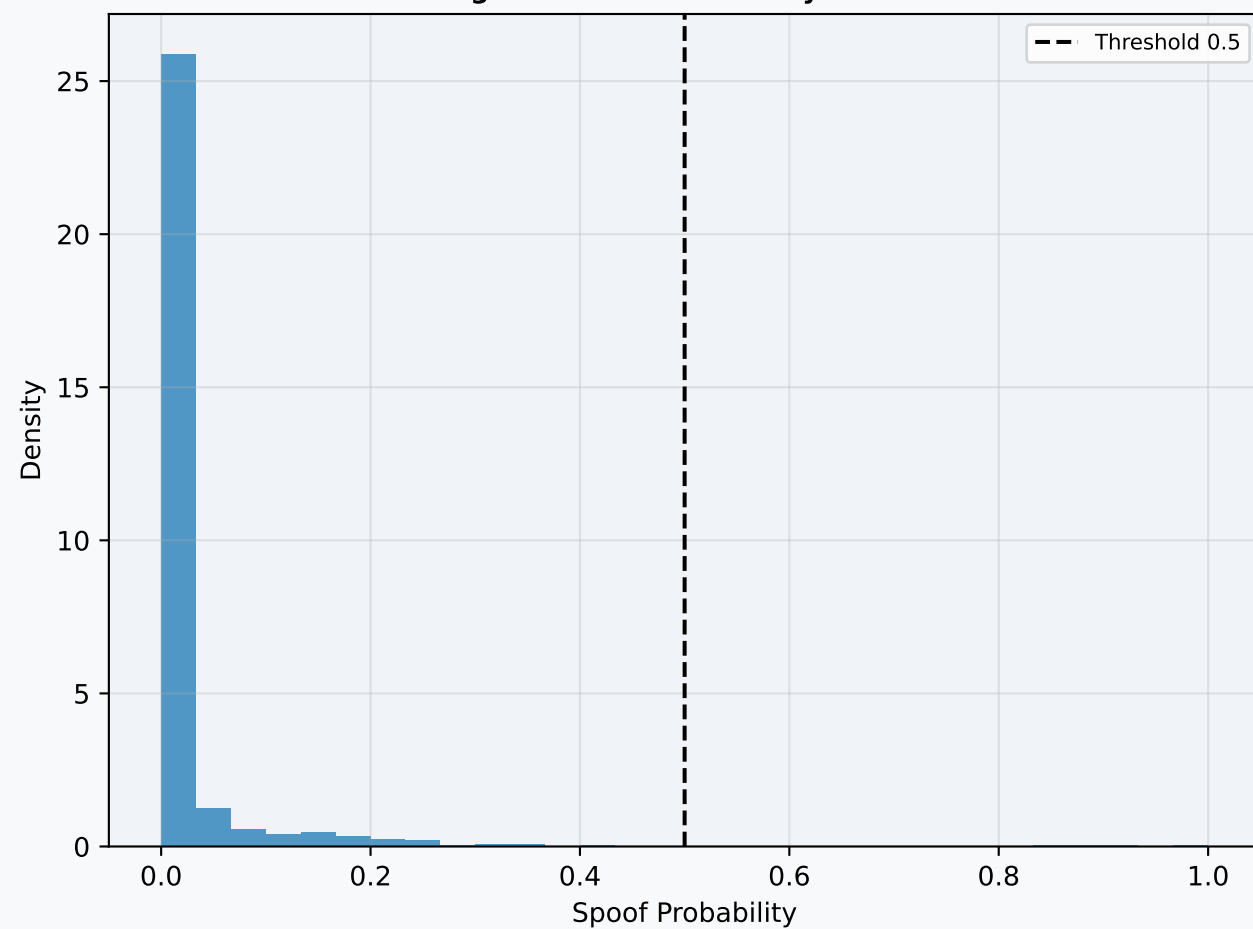


Blue bars = real audio sets | Red bars = spoof audio sets

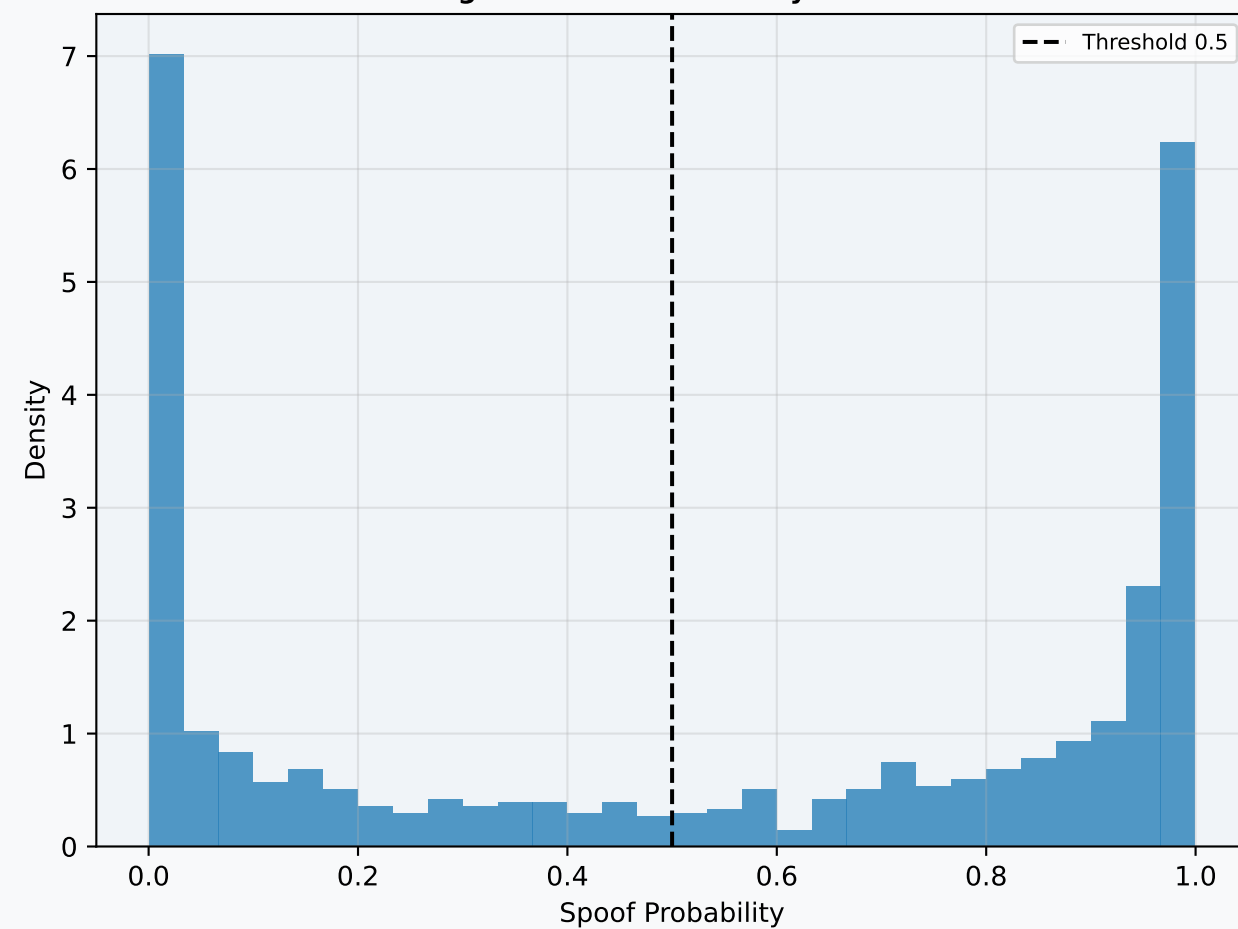
XGBoost Spoof Probability Distributions — All 6 Sets

(Real audio should cluster near 0, Spoof near 1)

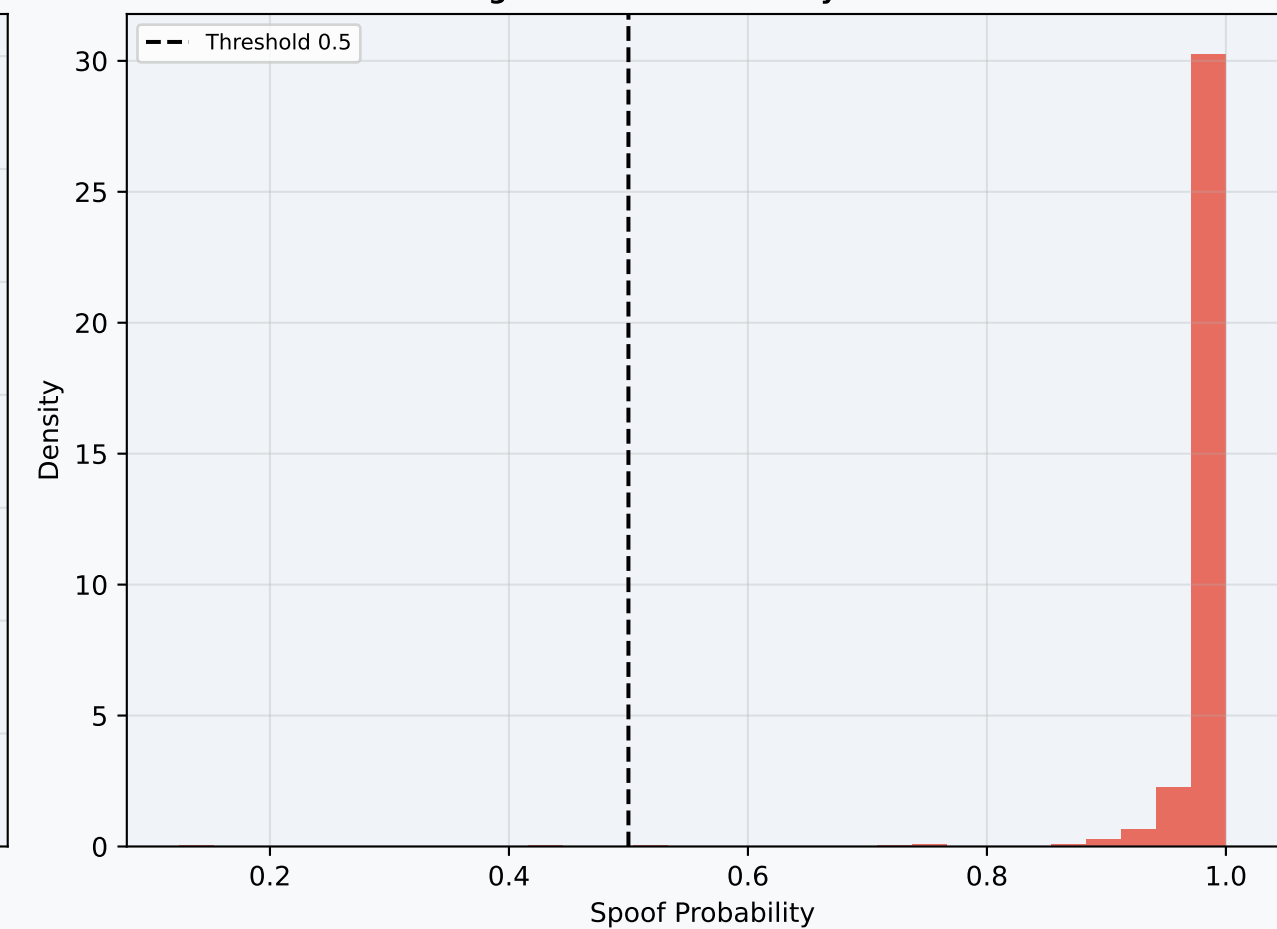
Set A — REAL (trained on)
avg=0.027 XGB accuracy=99.0%



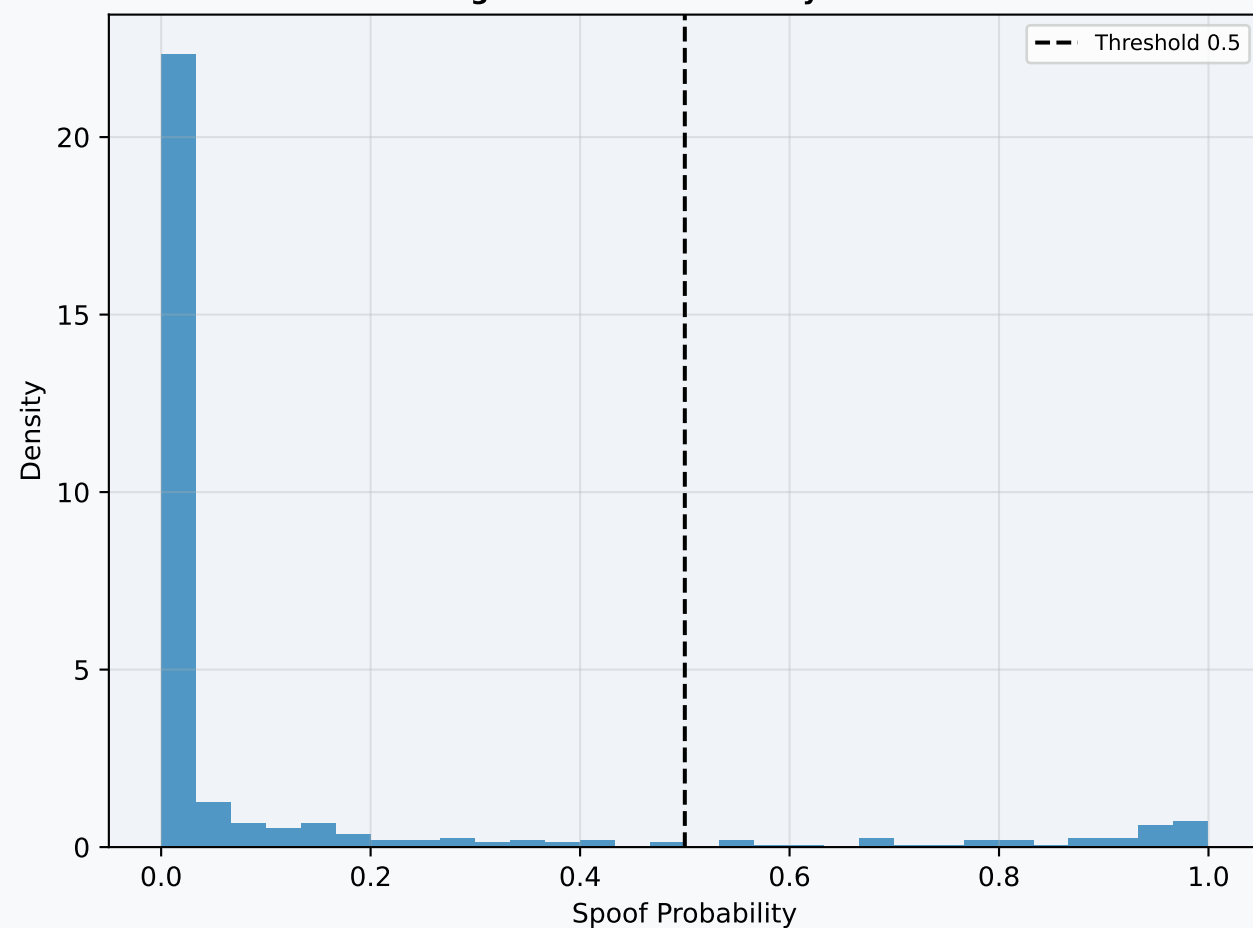
Set B — REAL (NOT trained on)
avg=0.525 XGB accuracy=46.1%



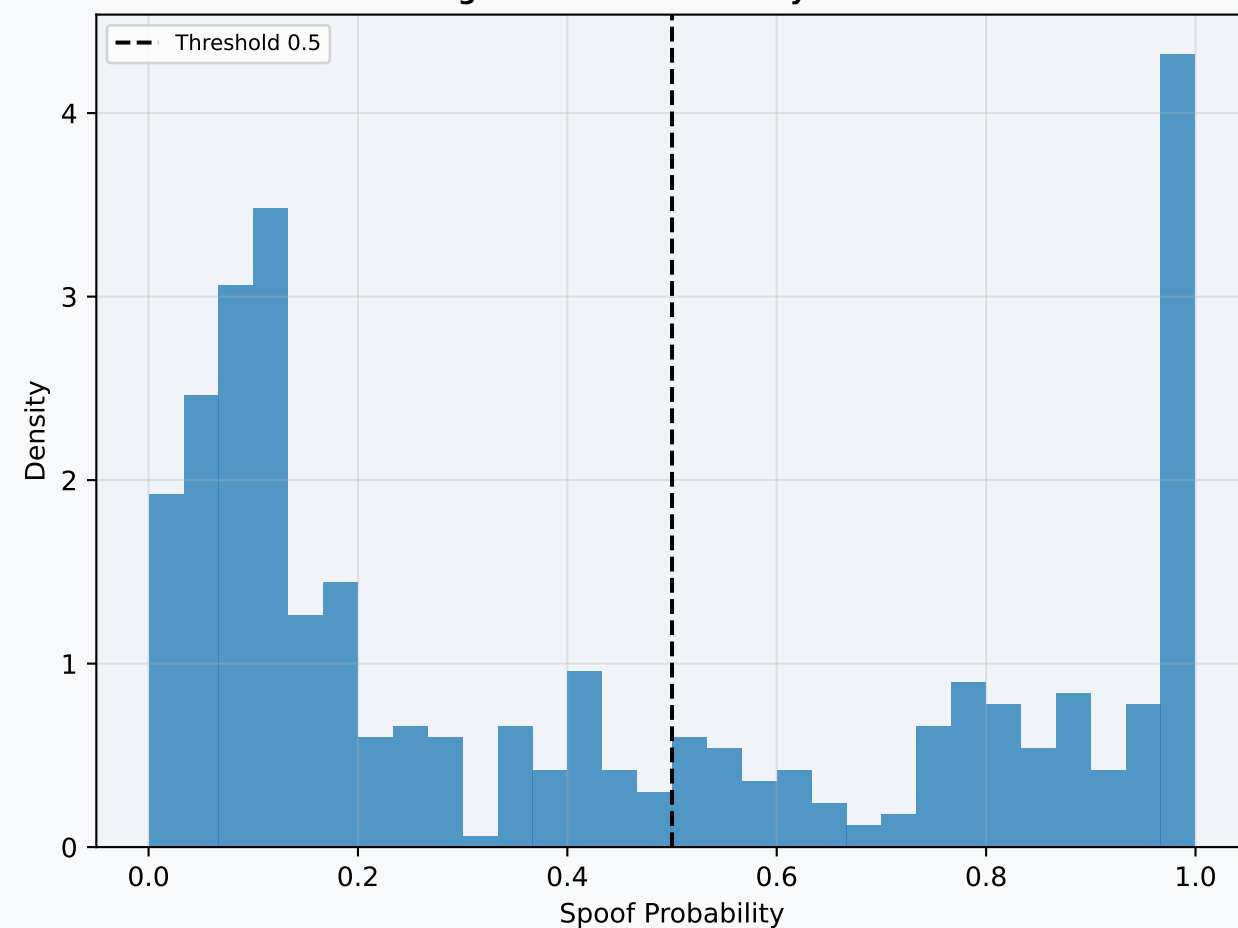
Set C — SPOOF (trained on)
avg=0.981 XGB accuracy=99.2%



Set D — REAL (NOT trained on)
avg=0.111 XGB accuracy=90.4%



Set E — REAL (NOT trained on)
avg=0.426 XGB accuracy=61.0%



Set F — SPOOF (NOT trained on)
avg=0.863 XGB accuracy=91.0%

